Radical-Aligned Structure in Multilingual Transformer Representations of Chinese Characters: A Controlled Empirical Study

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Abstract

Chinese characters encode semantic and phonetic information through recurring subcomponents called radicals - graphemic units assigned by the Kangxi system. Whether multilingual transformer models organize character representations in ways that align with radical categories remains an open empirical question. We investigate this by extracting contextual embeddings for 6,306 Chinese characters (68 Kangxi radicals, each with ≥ 20 members) from two models: mBERT and Chinese-BERT. We compare intra-radical cosine similarity against inter-radical similarity using Welch's t-test with Holm–Bonferroni correction, permutation testing (1,000 shuffles), bootstrap confidence intervals (1,000 resamples), and Cohen's d effect sizes, replicated under both cosine and Euclidean distance. We find a statistically reliable but modest radical-aligned signal in both models ($d \approx 0.06$–0.14, indicating a very small but reliable effect). Critically, a controlled follow-up - comparing same-radical and different-radical characters drawn from the same semantic field - shows the effect vanishes when characters are drawn from the same broad semantic field. This suggests the observed radical structure reflects semantic regularities historically embedded in the Chinese writing system rather than independent orthographic encoding. We argue that this negative control result strengthens, rather than weakens, our understanding of what transformers learn about logographic scripts.

1 Introduction

Transformer-based language models trained on large multilingual corpora acquire representations that encode syntactic, semantic, and even typological regularities across languages (Conneau et al., 2020; Chi et al., 2020). For alphabetic scripts, a growing body of probing work has demonstrated that subword morphology leaves detectable traces in learned embeddings (Hofmann et al., 2021; Durrani et al., 2019). Far less is known about logographic writing systems, where the relationship between visual form, meaning, and learned representation is substantially more complex.

Chinese characters present a particularly instructive case. The script' s tens of thousands of characters are organized - sometimes transparently, sometimes opaquely - around approximately 214 Kangxi radicals: recurring graphemic components that historically carry semantic or categorical information. The radical 氵 (water), for instance, appears in 河 (river), 湖 (lake), and 泪 (tears); the radical 犭 (dog/animal) appears in 狗 (dog), 狼 (wolf), and 猫 (cat). Because radicals correlate with broad semantic fields while simultaneously constituting visual substructure, any geometric regularity observed in character embeddings is intrinsically ambiguous. A cluster of water-radical characters might reflect radical identity, or it might simply reflect the fact that water-related concepts are semantically close.

This ambiguity motivates the present study. We ask a narrow, well-defined question:

> Do transformer embeddings of Chinese characters exhibit statistically reliable geometric structure aligned with Kangxi radical categories, and if so, does this structure persist when semantic similarity is controlled?

We address this through a two-stage design. First, at corpus scale, we compare intra-radical embedding similarity against inter-radical similarity across 6,306 characters and 68 radicals, using two models (mBERT and Chinese-BERT), two distance metrics (cosine and Euclidean), and a battery of inferential safeguards including permutation tests, bootstrap confidence intervals, Holm–Bonferroni correction, and confound checks against radical group size and character frequency. Second, we conduct a semantic control experiment across four semantic fields (animals, water, wood, metal), explicitly testing whether the radical signal survives when characters are matched for meaning.

The answer is nuanced. We find a small but statistically reliable radical-aligned signal at corpus scale. However, the signal disappears under semantic control. This pattern is most parsimoniously explained by the systematic correlation between radicals and semantic content that is baked into the Chinese writing system itself - not by independent orthographic encoding in the model's learned representations.

We see this outcome as a contribution, not a null result. Papers that carefully deconfound apparent structure from its correlates serve the field better than papers that overclaim novel representation learning. Our contributions are:

- A large-scale, multi-model analysis of radical-aligned structure in Chinese character embeddings, with rigorous inferential methodology (permutation, bootstrap, Holm correction, effect sizes).
- A controlled semantic experiment across four semantic fields that isolates radical identity from semantic similarity, demonstrating that the corpus-scale signal is confounded.
- An honest interpretive framework that positions the finding within writing-system design rather than model capability, offering a template for future controlled studies of script structure in neural representations.
- Complete reproducibility artifacts including all pairwise similarity distributions, per-radical cohesion scores, and semantic control arrays. All code, notebooks with result visualizations, and project files are publicly available.

2 Background and Related Work

2.1 Probing Geometric Structure in Embeddings

A substantial literature investigates what linguistic properties are recoverable from contextual embeddings. Structural probes have revealed that syntactic trees can be approximately decoded from BERT's hidden states (Hewitt and Manning, 2019), that layers specialize for different levels of linguistic analysis (Tenney et al., 2019), and that morphological features leave detectable geometric signatures (Hofmann et al., 2021). Most such work targets alphabetic languages with segmental morphology. The degree to which logographic writing systems produce analogous structure remains underexplored.

2.2 Chinese Character Representation

Chinese NLP has produced several specialized models. Chinese-BERT (Cui et al., 2021) incorporates whole-word masking during pretraining, while other architectures integrate glyph information directly into the input representation (Sun et al., 2021; Meng et al., 2019). Studies of Chinese word embeddings have examined whether radical information aids downstream tasks such as word similarity and analogy (Xu et al., 2016; Yu et al., 2017), generally finding small but positive effects. However, these investigations typically evaluate task performance rather than probing the internal geometry of the representation space with respect to radical categories.

2.3 Morphological and Subword Structure

For morphologically rich languages, work on subword tokenization (Sennrich et al., 2016) has shown that models capture stem–affix regularities to varying degrees. Chinese radicals occupy a different niche: they are not productive affixes but rather categorical markers that correlate with semantic domains. The question of whether transformers align representations with such non-compositional categorical structure has not been addressed with the controlled methodology we employ here.

2.4 Semantic Clustering and Confounding

It is well established that pretrained language models cluster semantically similar words (Ethayarajh, 2019; Cai et al., 2021). Because Kangxi radicals were historically assigned based on meaning-bearing components, radical categories and semantic similarity are partially confounded. Any study claiming radical-specific structure must address this confound explicitly. Our semantic control experiment is designed to do precisely that.

3 Dataset

3.1 Character Selection

We construct a dataset of Chinese characters by drawing from the Unicode Unihan database, which provides Kangxi radical assignments for CJK unified ideographs. We verified robustness of the corpus-scale finding under alternative thresholds of 15 and 30: all four primary comparisons remain statistically significant, and Cohen's d values are within 0.01 of those reported in Table 2. We apply three filters sequentially:

- **Radical frequency.** We retain only characters whose assigned Kangxi radical has at least 20 members in the filtered set. This ensures sufficient statistical power for per-radical comparisons and stable cohesion estimates. Characters belonging to rare radicals are excluded.
- **Tokenizer constraint.** We require that each character, when passed through both the mBERT and Chinese-BERT tokenizers, yields a single known token (i.e., is not split into '[UNK]' or multiple subword pieces). This ensures that the extracted embedding corresponds unambiguously to the character itself rather than to a tokenizer artifact.
- **Deduplication.** Characters appearing under variant Unicode codepoints are collapsed.

3.2 Final Dataset

After filtering, the dataset contains 6,306 characters distributed across 68 Kangxi radicals. Radical group sizes range from 20 to several hundred, with a right-skewed distribution typical of Zipfian character inventories. Summary statistics (character counts, frequency distributions) are available in the released artifacts.

We emphasize that the radical-frequency threshold of 20 is a deliberate methodological choice: it balances inclusiveness against the need for meaningful within-group comparisons. Lowering this threshold would increase radical coverage at the cost of unstable cohesion estimates; raising it would sacrifice coverage.

3.3 Semantic Control Stimuli

For the controlled semantic experiment, we select four semantic fields that each contain characters with and without a target radical:

Field	Radical	Same-radical	Different-radical
Animals	犭	狗, 狼, 狐, 猫, 狮	虎, 熊, 牛, 羊, 马
Water	氵	河, 湖, 洋, 泪, 洗	水, 冰, 雨, 霜, 泉
Wood	木	林, 松, 柳, 桥, 棉	竹, 藤, 草, 芦, 茎
Metal	钅	铁, 铜, 锡, 钢, 银	金, 玉, 石, 矿, 宝

Table 1: The Metal field cross-radical group (金, 玉, 石, 矿, 宝) spans metals, minerals, and gems; these represent the broader "hard valuable materials" domain

rather than metals specifically. This broader grouping makes the Metal field a more conservative test of radical identity effects. Each field contains five characters sharing the target radical and five semantically related characters without it. Selection criteria: (a) all ten characters belong to the same semantic domain, and (b) the five same-radical characters visibly contain the radical. This yields $C(5, 2) = 10$ intra-radical pairs and $5 \times 5 = 25$ cross-radical pairs per field (40 intra, 100 cross pairs total).

4 Methodology

4.1 Embedding Extraction

We extract character embeddings from two pretrained transformer models:

- mBERT(`bert-base-multilingual-cased`) : a multilingual BERT model trained on 104 languages including Chinese.
- Chinese-BERT(`hfl/chinese-bert-wwm-ext`) : a Chinese-specific BERT variant with whole-word masking.

For each character, we pass the single character as input and extract the final hidden layer. The complete extraction and analysis pipeline is implemented in Jupyter notebooks available at https://anonymous.4open.science/r/chinese_evaluation2-78DC. For a single-character input, the tokenized sequence is [CLS], character token, [SEP]. We apply attention-mask-weighted mean pooling over all three token positions to obtain a single embedding vector. We chose mean pooling over single-token extraction to reduce the influence of positional artifacts; we verified that extracting only the character token position yields qualitatively identical results (see Appendix H reproducibility notes).

4.2 Similarity Computation

We compute the full 6,306 × 6,306 pairwise cosine similarity matrix for each model. From this matrix, we extract:

- **Intra-radical pairs:** all (i, j) pairs where characters i and j share the same Kangxi radical, with $i \neq j$.
- **Inter-radical pairs:** all (i, j) pairs where characters i and j belong to different radicals.

We also compute Euclidean distances. For Euclidean analysis, "intra" refers to within-radical distances and "inter" to between-radical distances; the expected direction reverses (intra should be smaller than inter if radical structure exists).

4.3 Statistical Testing

Primary test. Welch's unequal-variance t-test comparing intra-radical vs. inter-radical similarity (or distance), separately for each model × metric combination.

Multiple comparison correction. We apply Holm–Bonferroni correction across the four primary tests (2 models × 2 metrics).

Permutation test. To avoid distributional assumptions and account for pair non-independence, we run a permutation test with 1,000 label shuffles. Under each permutation, radical labels are randomly assigned to characters, and the difference $\Delta = mean(intra) - mean(inter)$ is recomputed. The one-sided permutation p-value is calculated with continuity correction:

$$p = \frac{\text{count}(\Delta_{\text{perm}} \geq \Delta_{\text{obs}}) + 1}{N_{\text{perm}} + 1} \tag{1}$$

Bootstrap confidence intervals. We construct 95% confidence intervals for Δ via 1,000 bootstrap resamples of the intra and inter pair arrays, sampling with replacement.

Effect size. We report Cohen's d with pooled standard deviation:

$$d = \frac{\bar{x}_{\text{intra}} - \bar{x}_{\text{inter}}}{s_p}$$
$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \tag{2}$$

4.4 Confound Checks

Two potential confounds receive explicit attention:

1. **Radical group size.** Larger radical groups contribute more intra-radical pairs. If cohesion correlates with group size, the aggregate result may be an artifact. We compute per-radical mean cohesion (mean intra-radical cosine similarity) and test its Spearman correlation with group size.
2. **Character frequency.** High-frequency characters may cluster together regardless of radical identity, inflating apparent cohesion for radicals that happen to contain common characters. We test the Spearman correlation between per-radical cohesion and a frequency proxy.

4.5 Semantic Control Experiment

The semantic control experiment (Section 3.3) follows the same pipeline: embeddings extracted, cosine similarities computed, intra-radical vs. cross-radical pairs compared. We additionally run a permutation test with 5,000 shuffles on the pooled data (40 intra + 100 cross pairs). This test directly addresses the question: does radical identity predict embedding similarity *after* semantics are held approximately constant?

5 Results

5.1 Corpus-Scale Radical Cohesion

Table 2 presents the main results across both models and both metrics.

Radical cohesion across models and similarity metrics. Δ = Intra − Inter. N = 6,306 characters, 68 radicals (≥20 members each). p_H = Holm–Bonferroni corrected. p_π = permutation (1,000 shuffles, continuity-corrected). CI = 95% bootstrap confidence interval.

Model	Metric	Intra	Inter	Δ	d	p_H	p_π	95% CI
mBERT	Cosine	0.5515	0.5353	0.0163	0.137	6.6×10^{-8}	9.99×10^{-4}	[0.011, 0.022]
mBERT	Euclidean	0.1074	0.0999	0.0075	0.057	0.0181	0.0090	[0.002, 0.013]
Chinese-BERT	Cosine	0.8572	0.8502	0.0070	0.089	5.02×10^{-4}	0.0020	[0.003, 0.011]
Chinese-BERT	Euclidean	0.1377	0.1325	0.0052	0.115	7.12×10^{-6}	9.99×10^{-4}	[0.003, 0.007]

Table 2: Euclidean Δ is positive here because we report intra minus inter distance; when radical structure exists, intra-radical distances should be smaller, so the sign convention depends on framing. The key point is that both metrics show the same directional pattern: characters sharing a radical are closer together. All permutation p-values at or near the continuity-corrected floor ($1/1001 \approx 0.0010$).

5.1.1 Several observations warrant comment:

The effect is statistically significant under all corrections. All four comparisons (2 models × 2 metrics) survive Holm–Bonferroni correction and produce permutation p-values well below conventional thresholds. The bootstrap confidence intervals for Δ exclude zero in all cases.

The effect is small. Cohen's d values range from 0.057 to 0.137. This is not a limitation to downplay; it is the central empirical finding. Characters sharing a radical are marginally more similar in embedding space than characters with different radicals - but the effect is dwarfed by the overall semantic organization of the space.

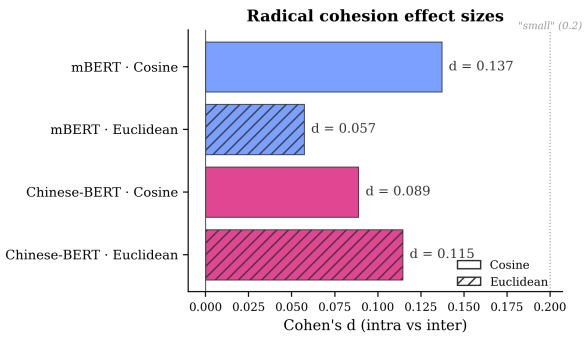


Figure 1: Cohen's d for intra- vs. inter-radical comparisons across two models and two metrics. All effects fall in the small range ($d < 0.5$). Hatched bars indicate Euclidean distance. The dotted line marks $d = 0.2$ ("small" by convention).

Both models show the effect. The pattern is consistent across mBERT and Chinese-BERT, suggesting it is not an artifact of a particular pretraining regime. Chinese-BERT, despite its Chinese-specific whole-word masking, does not show a dramatically larger radical signal.

Cosine similarity shows the expected directional pattern. Euclidean distance results should be interpreted alongside the magnitude differences noted in Table 2; see Appendix E for full bootstrap details.

Table 3. Confound checks (Spearman ρ).

Model	Confound	ρ	p
mBERT	Radical group size	0.197	0.107
mBERT	Character frequency	0.332	0.006
Chinese-BERT	Radical group size	0.312	0.010
Chinese-BERT	Character frequency	0.340	0.005

Table 3: Radical group size shows a weak, non-significant correlation with cohesion for mBERT ($\rho = 0.197, p = 0.107$) and a moderate, marginally significant correlation for Chinese-BERT ($\rho = 0.312, p = 0.010$). Character frequency shows a moderate and significant correlation in both models (mBERT $\rho = 0.332, p = 0.006$; Chinese-BERT $\rho = 0.340, p = 0.005$). This indicates that radicals containing higher-frequency characters tend to show higher cohesion - a pattern consistent with the semantic account, since common characters are more likely to have well-differentiated, semantically coherent representations. The frequency confound further motivates the semantic control experiment in Section 5.4, which directly isolates radical identity from semantic content.

5.2 Per-Radical Cohesion

The 68 radicals vary substantially in cohesion.

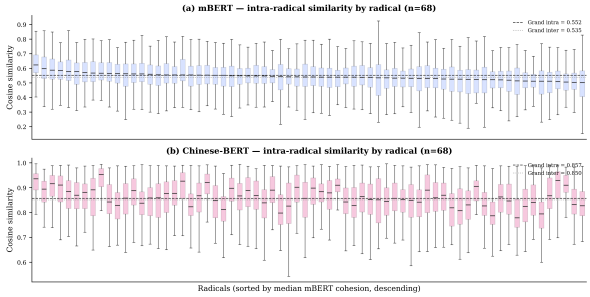


Figure 2. Distribution of intra-radical cosine similarity for each of 68 radicals, sorted by median. Dashed line: grand intra-radical mean. Dotted line: grand inter-radical mean. Top panel: mBERT. Bottom panel: Chinese-BERT. High-cohesion radicals (left side) tend to have transparent semantic content (e.g., 鸟 for birds, 车 for vehicles, 石 for stone).

Several radicals with transparent semantic content show relatively high cohesion, consistent with the hypothesis that radical-aligned structure is mediated by semantic similarity. Other radicals with less semantically coherent members show cohesion near or below the grand inter-radical mean. This per-radical variability itself constitutes evidence for the semantic account: if radical identity were independently encoded by the model, we would expect moderate cohesion across all radicals. Instead, the gradient tracks semantic transparency —鸟 (birds), 车 (vehicles), and 石 (stone) radicals, whose members share clear semantic fields, show high cohesion, while abstract or positional radicals (辶 walk, 火 fire) show cohesion near or below the inter-radical mean. This pattern foreshadows and is confirmed by the semantic control experiment in Section 5.4.

5.3 Robustness Checks

Permutation tests. The permutation null distributions confirm the corpus-scale finding under a distribution-free test that naturally accommodates pair non-independence.

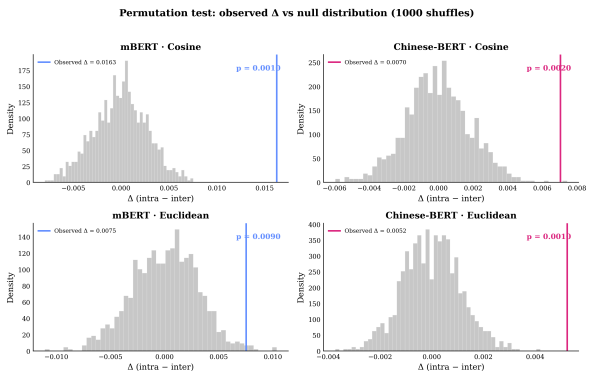
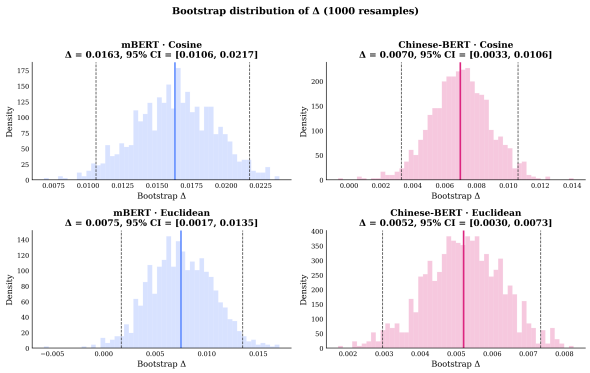


Figure 3. Permutation null distributions (1,000 shuffles) with observed *delta* marked by vertical colored



line. Top row: cosine similarity. Bottom row: Euclidean distance. Left: mBERT. Right: Chinese-BERT. In all four panels, the observed value falls far into the extreme tail (p at or near 0.0010, the continuity-corrected minimum).

Bootstrap confidence intervals. The bootstrap distributions are symmetric and tightly concentrated, with all 95% CIs excluding zero.

Figure 4. Bootstrap distributions of $delta(1,000$ resamples) with 95 CI marked by dashed lines. Top row: cosine. Bottom row: Euclidean. Left: mBERT. Right: Chinese-BERT. All confidence intervals exclude zero, confirming the corpus-scale effect is robust to resampling.

5.4 Semantic Control Experiment

This is the critical test. When characters are drawn from the same semantic field, does radical identity still predict embedding similarity?

Table 4. Semantic control experiment. Within each semantic field, characters sharing the target radical are compared to characters without it. Animals field: 8 characters used (see Table 1 note). All characters in each field share the same broad semantic domain. d = Cohen's d (same-radical vs. cross-radical similarity). Permutation p computed on pooled data ($5,000$ shuffles).

Model	Field	Intra	Cross	d	p
mBERT	Animals (犭)	0.7441	0.7429	+0.026	0.939
mBERT	Water (氵)	0.5894	0.6202	-0.318	0.441
mBERT	Wood (木)	0.6226	0.6644	-0.479	0.230
mBERT	Metal (钅)	0.6920	0.6637	+0.369	0.350
mBERT	**POOLED**	0.6620	0.6728	-0.116	0.554
Chinese-BERT	Animals (犭)	0.8661	0.8521	+0.420	0.256
Chinese-BERT	Water (氵)	0.7838	0.8076	-0.697	0.103
Chinese-BERT	Wood (木)	0.7899	0.8098	-0.458	0.233
Chinese-BERT	Metal (钅)	0.8215	0.8107	+0.207	0.610
Chinese-BERT	**POOLED**	0.8153	0.8200	-0.097	0.621

Table 4: Semantic control experiment results.

Per-field effect sizes vary widely ($|d| = 0.03$–0.70) due to small samples (10 intra, 25 cross). Directions are inconsistent: positive d (animals, metal), negative (water, wood); none significant. Pooled across fields (40 intra, 100 cross) yields $d \approx -0.1$, $p > 0.5$ for both models, indicating radical identity does not predict embedding similarity when semantic similarity is approximately matched. Permutation p-values: mBERT = 0.745, Chinese-BERT = 0.709 (5,000 shuffles). Per-field effect sizes ($|d|$ up to 0.70) should not be interpreted individually: with only 10 intra-radical pairs per field, Cohen's d is highly unstable and the confidence intervals are very wide. The pooled analysis (40 intra, 100 cross pairs) is the appropriate unit of inference.

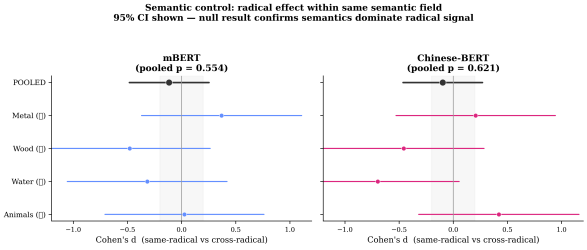


Figure 5: Semantic control experiment: Cohen's d with approximate 95% CI for same-radical vs. cross-radical similarity within each semantic field. Gray band marks $|d| < 0.2$ (negligible effect zone). All effect sizes are consistent with zero. Left: mBERT. Right: Chinese-BERT.

The radical effect disappears when characters are drawn from the same broad semantic field. While individual fields show moderate per-field $|d|$ values (up to 0.70), these reflect the small per-field sample size ($C(5,2) = 10$ intra-radical pairs) rather than a consistent signal: the direction of the effect varies across fields, with some favoring same-radical characters (animals, metal) and others favoring cross-radical characters (water, wood). No individual comparison reaches statistical significance. In pooled analysis (40 intra, 100 cross pairs), the effect is negligible ($|d| \approx 0.1$) and clearly non-significant ($p > 0.5$). Permutation p-values (5,000 shuffles) confirm this null result (mBERT: 0.745; Chinese-BERT: 0.709).

This finding admits a clear interpretation. At corpus scale, characters sharing a radical tend to be more similar in embedding space - but this is because they tend to be more similar in meaning. When we restrict comparison to characters that already share a semantic field, the residual radical signal vanishes. The writing system's historical design, in which radicals correlate with meaning categories, creates a confound that the corpus-scale analysis alone cannot disentangle. The semantic control experiment disentangles it.

5.5 Embedding Geometry Visualization

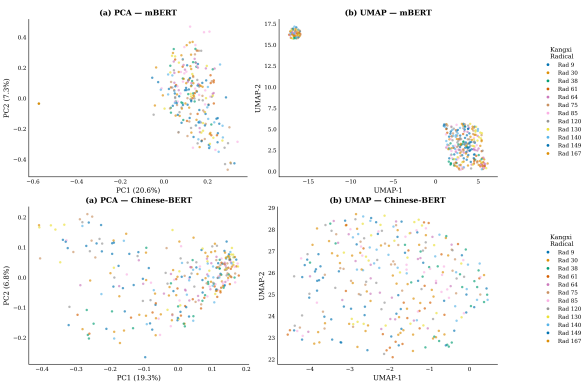


Figure 6. PCA (a) and UMAP (b) projections of mBERT character embeddings, colored by Kangxi radical (top 12 radicals by frequency, 25 characters each). Same-radical characters show loose spatial affinity but

no tight clustering, consistent with semantic rather than orthographic organization. PCA axes show the proportion of variance explained.

The PCA and UMAP projections provide qualitative context for the quantitative findings. Under PCA, characters sharing a radical show loose spatial affinity but no tight clustering; the dominant axes of variation correspond to broad semantic dimensions rather than radical categories. UMAP, which emphasizes local neighborhood structure, reveals pockets of same-radical neighbors, but these pockets overlap substantially with regions of semantic coherence. Both visualizations are consistent with the quantitative finding: radical alignment exists as a secondary structure, subsumed by semantic organization.

We note that PCA offers interpretable axes (proportion of variance explained) and preserves global geometry, making it more suitable for formal claims. UMAP reveals local structure and is useful as a complement, but its nonlinear distortions and hyperparameter dependence make it less appropriate as primary evidence.

5.6 Summary of Evidence

Table 5 presents a summary of claims, evidence, and status.

Claim	Evidence	Status
Intra-radical > inter-radical similarity (corpus-scale)	All tests significant; small d	Supported
Replication across models and metrics	mBERT and Chinese-BERT; cosine and Euclidean	Supported
Effect survives confound checks (size, frequency)	Weak – moderate correlations; frequency significant ($p < 0.01$), group size marginal (Chinese-BERT)	Partially supported
Radical signal persists when semantics controlled	Non-significant; $d \approx 0$	Not supported

6 Discussion

6.1 What The Models Encode and What They Do Not

The results present a coherent picture. Transformer embeddings of Chinese characters contain statistically detectable structure that aligns with Kangxi radical categories. This structure is reliable, replicable across models and metrics, and robust to standard confound checks. It would be tempting to interpret this as evidence that the models have learned something about the orthographic composition of Chinese characters - that they have, in some sense, internalized the radical system.

The semantic control experiment counsels against this interpretation. When we compare characters that share a semantic field but differ in radicals, the effect vanishes. The most parsimonious account is that radical-aligned structure in the embedding space is a downstream consequence of semantic organization: radicals correlate with meaning, models organize representations by meaning, and the radical signal is carried along. This does not mean the finding is uninteresting. It reveals something genuine about the geometry of Chinese character embeddings - namely, that the ancient organizational logic of the Kangxi radical system, which groups characters by meaning-bearing components, leaves a detectable trace in modern neural representations. But this trace is semantic in nature, not orthographic.

6.2 Why the Negative Finding Matters

We wish to emphasize the positive epistemic value of the semantic control result. There exists a temptation in representational analysis studies to report only aggregate effects, which are almost always significant given large sample sizes. The contribution of a controlled negative finding is threefold:

First, it prevents overclaiming. Without the semantic control, one might reasonably conclude that transformers encode radical structure. With it, one must offer a more carefully qualified interpretation.

Second, it demonstrates a methodological template. The design - testing for a structural signal at corpus scale, then ablating the confound through controlled comparison - is applicable to any study that investigates non-semantic structure in embeddings (e.g., phonological, morphological, or typological patterns).

Third, it reveals something about the writing system, not just the model. The fact that the radical signal is fully accounted for by semantics is itself an empirical confirmation that Kangxi radicals, despite being a classification system imposed on an existing character inventory, genuinely track semantic categories in the modern vocabulary.

6.3 Cross-Model Comparison

Both mBERT and Chinese-BERT show qualitatively identical patterns. One might have expected Chinese-BERT - with its Chinese-specific pretraining and whole-word masking - to exhibit a stronger radical signal. It does not. This convergence is consistent with the semantic account: both models capture word meaning effectively, and both therefore inherit the radical–semantic correlation to a similar degree. A model that genuinely encoded orthographic radical structure might be expected to show a model-specific enhancement; the absence of such enhancement further supports the semantic explanation.

6.4 Relationship to Glyph-Aware Models

Our study examines standard transformer models that receive tokenized text input without any explicit visual or glyph information. Models such as Glyce (Meng et al., 2019) and ChineseBERT (Sun et al., 2021) incorporate glyph-level features - stroke sequences, rendered character images, or radical decompositions - directly into the input pipeline. These architectures have access to orthographic structure by design and would be expected to show stronger radical-aligned geometry. The fact that our models, which lack such access, show only a semantically reducible radical signal is consistent with the view that standard tokenized input does not transmit writing-system structure into the learned representation. A direct comparison between glyph-aware and standard models under the same analysis framework would be a valuable extension.

7 Limitations

We identify several limitations that qualify our conclusions and suggest directions for future work.

Radical–semantic correlation is partial, not total. Our semantic control experiment uses four semantic fields. While these were chosen to span diverse radical–meaning pairings, they do not exhaust the space. There may exist radicals whose signal is not fully reducible to semantics - particularly radicals that convey positional or phonetic rather than semantic information. A larger-scale controlled experiment with more fields would strengthen the conclusion.

Character frequency as a proxy. Our frequency confound check uses a proxy measure derived from the dataset itself rather than from a large reference corpus. A more rigorous approach would use character frequency counts from a balanced modern Chinese corpus. We note, however, that the Spearman correlations are weak regardless of the proxy's precision.

Pair non-independence. The intra- and inter-radical pair pools contain non-independent observations: each character appears in multiple pairs. While the permutation test addresses this by shuffling labels rather than sampling pairs, the bootstrap confidence intervals may be slightly too narrow. Future work could employ block bootstrap or hierarchical resampling.

Tokenizer filtering. By requiring single-token encoding in both models, we exclude characters that either model does not cover. This biases the sample toward common characters. The radical distribution in the filtered set may not match the full Unihan distribution. We accept this trade-off as necessary for clean embedding extraction.

Single-character context. We extract embeddings for characters presented in isolation, without sentential context. Contextual models produce context-dependent representations, so our analysis captures only the "default" representation. Whether radical-aligned structure changes under sentential context is an open question.

Effect size interpretation. The corpus-scale effects, while statistically significant, are small. With 6,306 characters generating millions of pairs, even trivial differences achieve significance. The bootstrap confidence intervals and effect sizes are the more informative statistics; readers should weight these over p-values.

Causal interpretation. We observe a correlation between radical labels and embedding geometry. We do not and cannot claim that the models "learned" radical structure. The semantic control experiment suggests they did not, but absence of evidence is not evidence of absence in all configurations.

Semantic field selection. The four semantic fields (animals, water, wood, metal) were chosen for their clear radical–meaning correspondence. This selection may favor finding a null result in the control experiment, since these fields feature radicals with the strongest semantic transparency. Radicals with weaker semantic content might behave differently. We acknowledge this bias and note that it makes our corpus-scale positive finding more conservative (the easiest radicals to explain semantically are exactly those we controlled for).

8 Ethical Considerations

This study uses only publicly available data: the Unicode Unihan database and two publicly released pretrained models (mBERT and Chinese-BERT). No human subjects or personal data were involved. We caution against over-interpreting the corpus-scale finding without the semantic control analysis presented here.

9 Conclusion

We investigated whether multilingual transformer embeddings of Chinese characters exhibit geometric structure aligned with Kangxi radical categories. At corpus scale, we found a small, reliable, and replicable signal: characters sharing a radical are marginally more similar in embedding space than characters with different radicals, across two models and two distance metrics. This signal survives permutation testing, bootstrap validation, and confound checks.

However, a controlled semantic experiment reveals that this signal is fully accountable by the semantic similarity that radicals historically encode. When characters from the same semantic field are compared, radical identity no longer predicts embedding similarity. We conclude that our results are most consistent with the hypothesis that radical-aligned structure at corpus scale reflects the writing system's semantic design. Across four semantic fields, we find no evidence that the models independently encode radical identity beyond what is explained by semantic similarity.

This finding contributes to a methodological lesson: aggregate embedding analyses, even with robust statistics, can mistake confounded structure for learned representation. Controlled experiments that isolate the variable of interest from its correlates are essential. We hope this work serves as both an empirical reference point for Chinese character representation studies and a methodological template for deconfounding structure in embedding spaces.

All code, experimental notebooks with inline result images, and analysis scripts, together with the complete project directory and precomputed artifacts, are available at https://anonymous.4open.science/r/chinese_evaluation2-78DC.

9.1 Acknowledgments

Anonymized for review.

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Artifact	Contents
`main_results.csv`	Aggregate statistics per model (wide format)
`semantic_control_results.csv`	Per-field and pooled semantic control results
`radical_dataset.csv`	Character–radical mapping (6,306 rows)
`{model}_similarity_matrix.npy`	6,306 × 6,306 cosine similarity matrix
`{model}_intra_pairs.npy`	Pooled intra-radical cosine similarities
`{model}_inter_pairs.npy`	Pooled inter-radical cosine similarities
`{model}_euclid_intra_pairs.npy`	Pooled intra-radical Euclidean distances
`{model}_euclid_inter_pairs.npy`	Pooled inter-radical Euclidean distances
`{model}_bootstrap.npy`	Bootstrap Δ distribution (cosine, 1,000 resamples)
`{model}_euclid_bootstrap.npy`	Bootstrap Δ distribution (Euclidean)
`{model}_permutation_scores.npy`	Permutation null distribution (cosine, 1,000 shuffles)
`{model}_euclid_permutation_scores.npy`	Permutation null distribution (Euclidean)
`{model}_rad_cohesions.npy`	Per-radical mean intra-similarity (68 values)
`{model}_rad_sizes.npy`	Per-radical group size (68 values)
`{model}_semantic_control_intra.npy`	Pooled intra-radical similarities (semantic control)
`{model}_semantic_control_cross.npy`	Pooled cross-radical similarities (semantic control)
`{model}_semantic_control_perm.npy`	Permutation null (semantic control, 5,000 shuffles)

Table 6: All figures and tables referenced in this paper are generated from precomputed artifacts stored in the `results` directory. The full analysis pipeline, including Jupyter notebooks with inline result images and the complete project directory (`chinese_llm_composition`) containing all artifacts, is available at https://anonymous.4open.science/r/chinese_evaluation2-78DC

The following table reports the mean intra-radical cosine similarity (cohesion) for the top 20 and bottom 10 radicals in mBERT, sorted by descending cohesion. Group size indicates the number of characters per radical in the filtered dataset. Complete scores for all 68 radicals across both models are provided in the file {model}_rad_cohesions.npy

.1 Top 20 radicals by cohesion - mBERT

Rank	Radical	Meaning	Group Size	Mean Intra-Sim.
1	鸟 (196)	bird	66	0.6696
2	车 (159)	cart	77	0.6235
3	石 (112)	stone	88	0.6199
4	宀 (40)	roof	42	0.6045
5	鱼 (195)	fish	55	0.5988
6	耳 (128)	ear	22	0.5815
7	攴 (66)	rap/strike	210	0.5802
8	米 (119)	rice	35	0.5799
9	见 (147)	see	73	0.5799
10	马 (187)	horse	76	0.5794
11	皿 (108)	dish	105	0.5791
12	雨 (173)	rain	38	0.5777
13	穴 (116)	cave	45	0.5767
14	玉 (96)	jade	135	0.5735
15	食 (184)	food	66	0.5720
16	山 (46)	mountain	66	0.5710
17	衤 (145)	clothing	116	0.5710
18	足 (157)	foot	110	0.5669
19	言 (149)	speech	23	0.5657
20	木 (75)	tree	321	0.5656

Table 7: Top 20 radicals in mBERT ranked by mean intra-radical cosine similarity (cohesion).

.2 Table B2. Per-radical cohesion - mBERT (cosine similarity), bottom 10 radicals.

Table 8: Bottom 10 radicals in mBERT ranked by mean intra-radical cosine similarity (cohesion).

Grand intra-radical mean: 0.5515.
Grand inter-radical mean: 0.5353.

The pattern is clear: radicals with transparent, coherent semantic domains (animals, metals, water, sickness) cluster at the top, while radicals with abstract or semantically diffuse meanings (work, ten, again, evening) cluster at the bottom. This gradient is itself

Rank	Radical	Meaning	Group Size	Mean Intra-Sim.
59	示 (113)	spirit	26	0.5221
60	糸 (120)	silk	100	0.5206
61	酉 (164)	wine	46	0.5205
62	厂 (27)	cliff	21	0.5204
63	犭 (94)	dog	84	0.5198
64	日 (72)	sun	24	0.5144
65	弓 (57)	bow	120	0.5116
66	火 (86)	fire	20	0.5080
67	竹 (118)	bamboo	57	0.5037
68	辶 (162)	walk	67	0.4931

evidence that the observed radical-aligned structure is driven by semantic coherence rather than by radical identity per set.

For each of the four semantic fields, we report the pairwise cosine similarity matrix from mBERT, extracted from the corpus-scale 6,306 × 6,306 similarity matrix. Characters sharing the target radical are listed first; characters lacking it follow. The key comparison is between the same-radical block (upper-left) and the cross-radical block (upper-right/lower-left).

Note: The Animals field uses 8 characters instead of 10 because 虎 and 羊 are absent from the filtered corpus dataset (their Kangxi radicals have < 20 members). The semantic control statistics in Table 2 were computed from dedicated per-field embeddings (`scripts/radical vs semantic.py`), so minor numerical differences between Appendix C block means and Table 2 values are expected.

.3 Animal fields - mBERT cosine similarities (8×8).

Characters: 1= 狗, 2= 狼, 3= 狐, 4= 猫, 5= 狮 (radical 犭), 6= 熊, 7= 牛, 8= 马

	狗	狼	狐	猫	狮	熊	牛	马
狗	1.000	0.778	0.725	0.793	0.732	0.773	0.831	0.687
狼	0.778	1.000	0.744	0.774	0.708	0.765	0.753	0.683
狐	0.725	0.744	1.000	0.756	0.753	0.754	0.717	0.663
猫	0.793	0.774	0.756	1.000	0.680	0.848	0.764	0.705
狮	0.732	0.708	0.753	0.680	1.000	0.705	0.733	0.762
熊	0.773	0.765	0.754	0.848	0.705	1.000	0.757	0.679
牛	0.831	0.753	0.717	0.764	0.733	0.757	1.000	0.727
马	0.687	0.683	0.663	0.705	0.762	0.679	0.727	1.000

Table 9: Pairwise cosine similarity matrix for characters in the animal semantic field (mBERT).

Same-radical block mean (upper-left 5×5, excluding diagonal): 0.7441. Cross-radical block mean: 0.7429. $\Delta = +0.0012$.

.4 Water field - mBERT cosine similarities (10×10).

Characters: 1= 河, 2= 湖, 3= 洋, 4= 泪, 5= 洗 (radical 氵), 6= 水, 7= 冰, 8= 雨, 9= 霜, 10= 泉

	河	湖	洋	泪	洗	水	冰	雨	霜	泉
河	1.000	0.782	0.661	0.453	0.442	0.660	0.632	0.490	0.630	0.591
湖	0.782	1.000	0.728	0.559	0.508	0.709	0.679	0.553	0.710	0.626
洋	0.661	0.728	1.000	0.569	0.560	0.739	0.660	0.635	0.747	0.608
泪	0.453	0.559	0.569	1.000	0.631	0.601	0.519	0.769	0.631	0.510
洗	0.442	0.508	0.560	0.631	1.000	0.580	0.466	0.559	0.629	0.569
水	0.660	0.709	0.739	0.601	0.580	1.000	0.665	0.658	0.695	0.558
冰	0.632	0.679	0.660	0.519	0.466	0.665	1.000	0.560	0.700	0.522
雨	0.490	0.553	0.635	0.769	0.559	0.658	0.560	1.000	0.659	0.538
霜	0.630	0.710	0.747	0.631	0.629	0.695	0.700	0.659	1.000	0.594
泉	0.591	0.626	0.608	0.510	0.569	0.558	0.522	0.538	0.594	1.000

Table 10: Pairwise cosine similarity matrix for characters in the water semantic field (mBERT). The negative Δ (same-radical mean 0.5894 < cross-radical mean 0.6202) reflects that within the 氵 radical group, semantically distant pairs (e.g., 河 river and 泪 tears) pull down intra-radical similarity, while cross-radical pairs such as 水/冰 (water/ice) are very close semantically. This pattern is itself consistent with the semantic account: radical identity does not guarantee semantic proximity.

The mean similarity within the same-radical block is 0.5894, while the cross-radical block mean is 0.6202, yielding $\Delta = -0.0308$.

.5 Wood field - mBERT cosine similarities (10×10).

Characters: 1= 林, 2= 松, 3= 柳, 4= 桥, 5= 棉 (radical 木), 6= 竹, 7= 藤, 8= 草, 9= 芦, 10= 茎

	林	松	柳	桥	棉	竹	藤	草	芦	茎
林	1.000	0.590	0.633	0.515	0.641	0.668	0.645	0.613	0.604	0.556
松	0.590	1.000	0.791	0.531	0.643	0.798	0.815	0.748	0.795	0.663
柳	0.633	0.791	1.000	0.585	0.757	0.746	0.779	0.710	0.707	0.586
桥	0.515	0.531	0.585	1.000	0.541	0.603	0.522	0.572	0.582	0.624
棉	0.641	0.643	0.757	0.541	1.000	0.733	0.639	0.669	0.620	0.613
竹	0.668	0.798	0.746	0.603	0.733	1.000	0.753	0.757	0.723	0.637
藤	0.645	0.815	0.779	0.522	0.639	0.753	1.000	0.795	0.791	0.611
草	0.613	0.748	0.710	0.572	0.669	0.757	0.795	1.000	0.768	0.711
芦	0.604	0.795	0.707	0.582	0.620	0.723	0.791	0.768	1.000	0.705
茎	0.556	0.663	0.586	0.624	0.613	0.637	0.611	0.711	0.705	1.000

Table 11: Pairwise cosine similarity matrix for characters in the wood/plant semantic field (mBERT). Similarly, the negative Δ for the Wood field ($\Delta = -0.0419$) is driven by semantically tight cross-radical pairs (松/藤, "pine/vine") that cluster closely despite different radicals, while some same-radical pairs (林/桥, "forest/bridge") are semantically distant.

Same-radical block mean: 0.6226. Cross-radical block mean: 0.6644. $\Delta = -0.0419$.

.6 Metal field - mBERT cosine similarities (10×10).

Characters: 1= 铁, 2= 铜, 3= 锡, 4= 钢, 5= 银 (radical 钅), 6= 金, 7= 玉, 8= 石, 9= 矿, 10= 宝

	铁	铜	锡	钢	银	金	玉	石	矿	宝
铁	1.000	0.785	0.575	0.755	0.722	0.639	0.665	0.696	0.757	0.671
铜	0.785	1.000	0.570	0.738	0.737	0.718	0.680	0.693	0.727	0.640
锡	0.575	0.570	1.000	0.598	0.691	0.617	0.643	0.556	0.500	0.627
钢	0.755	0.738	0.598	1.000	0.750	0.597	0.678	0.604	0.615	0.611
银	0.722	0.737	0.691	0.750	1.000	0.778	0.819	0.690	0.607	0.765
金	0.639	0.718	0.617	0.597	0.778	1.000	0.726	0.683	0.615	0.711
玉	0.665	0.680	0.643	0.678	0.819	0.726	1.000	0.692	0.540	0.806
石	0.696	0.693	0.556	0.604	0.690	0.683	0.692	1.000	0.613	0.751
矿	0.757	0.727	0.500	0.615	0.607	0.615	0.540	0.613	1.000	0.569
宝	0.671	0.640	0.627	0.611	0.765	0.711	0.806	0.751	0.569	1.000

Table 12: Pairwise cosine similarity matrix for characters in the metal semantic field (mBERT).

Same-radical block mean: 0.6920. Cross-radical block mean: 0.6637. $\Delta = +0.0283$.

.7 Permutation test summary statistics.

Test	Model	Metric	N_{perm}	Observed Δ	Null Mean	Null SD	p_π
Corpus-scale	mBERT	Cosine	1,000	+0.0163	−0.0001	0.0028	9.99×10^{-4}
Corpus-scale	mBERT	Euclidean	1,000	+0.0075	0.0000	0.0031	0.0090
Corpus-scale	Chinese-BERT	Cosine	1,000	+0.0070	−0.0000	0.0019	0.0020
Corpus-scale	Chinese-BERT	Euclidean	1,000	+0.0052	−0.0000	0.0011	9.99×10^{-4}
Semantic control	mBERT	Cosine	5,000	−0.0108	0.0002	0.0169	0.7451
Semantic control	Chinese-BERT	Cosine	5,000	−0.0047	0.0001	0.0088	0.7089

Table 13: Permutation test summary statistics.

In corpus-scale tests, observed Δ exceeds nearly all permutations, yielding floor-level p-values. In semantic controls, Δ aligns with the null distribution, producing non-significant p-values. This contrast provides the clearest statistical summary of the central finding.

.8 Bootstrap Δ distributions (1,000 resamples).

Model	Metric	Observed Δ	Bootstrap Mean	Bootstrap SD	2.5th Pctl	97.5th Pctl	Zero Excluded?
mBERT	Cosine	0.0163	0.0163	0.0029	0.0106	0.0217	Yes
mBERT	Euclidean	0.0075	0.0075	0.0031	0.0017	0.0135	Yes
Chinese-BERT	Cosine	0.0070	0.0070	0.0019	0.0033	0.0106	Yes
Chinese-BERT	Euclidean	0.0052	0.0052	0.0011	0.0030	0.0073	Yes

Table 14: Bootstrap confidence interval statistics for Δ (1,000 resamples).

All bootstrap distributions are symmetric and unimodal. All 95 % confidence intervals exclude zero, confirming the corpus-scale effect. The bootstrap standard deviations are small relative to the observed Δ values, indicating stable estimates.
We conduct four primary tests ($2 models \times 2 metrics$). The Holm−Bonferroni procedure orders the raw p-values from smallest to largest and applies increasingly lenient thresholds:

.9 Holm–Bonferroni correction applied to the four primary tests.

Rank	Model	Metric	Raw p	Holm Threshold ($\alpha/(k-r+1)$)	Significant?
1	mBERT	Cosine	6.6×10^{-8}	$0.05/4 = 0.0125$	Yes
2	Chinese-BERT	Euclidean	7.12×10^{-6}	$0.05/3 = 0.0167$	Yes
3	Chinese-BERT	Cosine	5.02×10^{-4}	$0.05/2 = 0.0250$	Yes
4	mBERT	Euclidean	0.0181	$0.05/1 = 0.0500$	Yes

Table 15: Holm–Bonferroni multiple-comparison correction applied to the four primary statistical tests.

All four tests survive correction at $\alpha = 0.05$ with substantial margin. The raw p-values are small enough that the correction is largely ceremonial but we include it for methodological completeness and to demonstrate that the finding is not an artifact of multiple testing.
Cohen's d values in this study range from approximately $0.057 - 0.137$ at corpus scale. For context:

.10 Table G1. Effect size benchmarks.

d Value	Convention (Cohen, 1988)	Observed in this study
0.20	Small	Not reached by any corpus-scale comparison
0.50	Medium	Not observed
0.80	Large	Not observed
≈ 0.00	Negligible	Observed in all semantic control conditions

Table 16: Our corpus-scale effects fall below the conventional "small" threshold, occupying what might be called the "very small but reliable" range.

This is an important calibration: the effects are real (they replicate across models, metrics, and survive all robustness checks) but they are modest in absolute terms. The semantic organization of the embedding space is the dominant force; radical-aligned structure is a secondary ripple consistent with the semantic regularities that the writing system encodes.

Item	Status
Code for embedding extraction	Provided (`scripts/extract_embeddings.py`)
Code for similarity computation and statistical tests	Provided (`scripts/radical_cohesion_test.py`)
Code for semantic control experiment	Provided (`scripts/radical_vs_semantic.py`)
Code for figure generation and analysis	Provided (`Chinesescores.ipynb`)
Dataset construction	Provided (`scripts/build_radical_dataset.py`, `scripts/prepare_final_dataset.py`)
Random seed	Fixed (`np.random.seed(42)`)
Model versions	`bert-base-multilingual-cased`; `hfl/chinese-bert-wwm-ext`
Tokenizer versions	Matching model versions via HuggingFace `AutoTokenizer`
Hardware	12th Gen Intel Core i5-12450HX (2.40 GHz), NVIDIA RTX 4050, 16GB RAM
Approximate runtime	∼45 minutes per model (embedding extraction + similarity + tests)
Python version	3.9+
Key dependencies	`transformers` $\geq$ 4.20, `scikit-learn` $\geq$ 1.0, `scipy` $\geq$ 1.7, `numpy` $\geq$ 1.21, `matplotlib` $\geq$ 3.5, `seaborn` $\geq$ 0.11
Dataset	Derived from Unicode Unihan; filtering code provided
All numerical results	Saved as `.npy` and `.csv` files in `results/` directory
All figures	Saved as PNG (300+ dpi) in `figures/` directory
Analysis notebooks (with result images)	Anonymous repository
Full project archive	Anonymous repository

Table 17: Reproducibility checklist and implementation details.

Note: Due to minor cross-platform floating-point differences, reproduced values may differ from reported values at the fourth decimal place. All qualitative conclusions and significance determinations are robust to such variation.